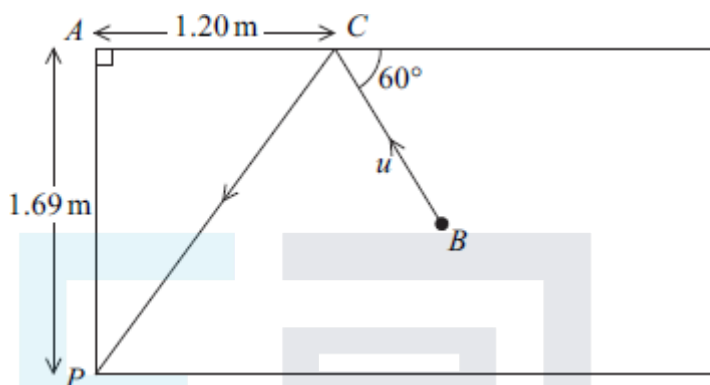


Q1.

The diagram shows part of a horizontal snooker table of width 1.69 m.

A player strikes the ball B directly, and it moves in a straight line. The ball hits the cushion of the table at C before rebounding and moving to the pocket at P at the corner of the table, as shown in the diagram. The point C is 1.20 m from the corner A of the table. The ball has mass 0.15 kg and, immediately before the collision with the cushion, it has velocity u in a direction inclined at 60° to the cushion. The **table** and the **cushion** are modelled as smooth.



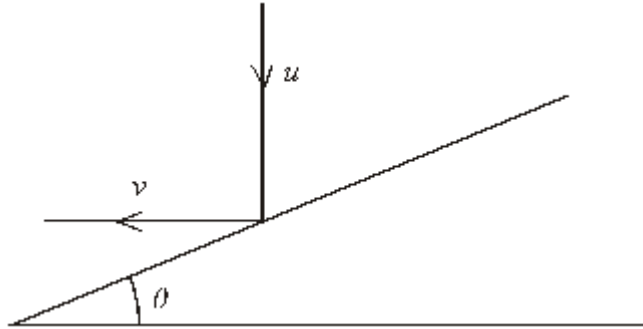
- (a) Find the coefficient of restitution between the ball and the cushion. (5)
- (b) Show that the magnitude of the impulse on the cushion at C is approximately $0.236u$. (4)
- (c) Find, in terms of u , the time taken between the ball hitting the cushion at C and entering the pocket at P . (3)
- (d) Explain how you have used the assumption that the cushion is smooth in your answers. (1)

(Total 13 marks)

Q2.

A small smooth ball B , of mass m , falls vertically and collides with a fixed smooth plane inclined at an angle of θ to the horizontal. The speed of B just before the collision is u . Immediately after the collision, B moves horizontally with speed v .

The diagram shows the velocities of B immediately before and after the collision.



- (a) Briefly explain why the component of the velocity of B parallel to the plane is not changed by the collision.

(2)

- (b) Given that the coefficient of restitution between B and the plane is $\frac{3}{4}$, show that

$$3u \cos \theta = 4v \sin \theta$$

(2)

- (c) Hence show that the value of θ is approximately 40.9° .

(4)

- (d) Show that the magnitude of the impulse exerted by the plane on B is approximately $1.32 mu$.

(5)

(Total 13 marks)

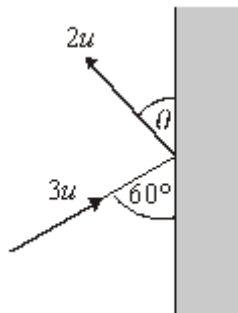
Q3.

A sphere of mass m , moving on a smooth horizontal surface, hits a smooth vertical wall. Just before it hits the wall, the sphere is moving at an angle of 60° to the wall with speed $3u$.

©2025 Exam Papers Practice. All rights reserved.

After the collision, the sphere has speed $2u$ and is moving at an angle of θ to the wall.

The diagram shows the situation viewed from above.



Model the sphere as a particle.

- (a) (i) Show that $\cos \theta = \frac{3}{4}$.

- (3)
- (ii) Find the coefficient of restitution between the wall and the sphere. (3)
- (iii) Find the angle through which the direction of the sphere is changed. (2)
- (b) Find, in terms of u and m , the impulse exerted on the sphere by the wall during this collision. (3)
- (Total 11 marks)**

Q4.

A tennis ball, of mass m , is hit towards a smooth vertical wall.

Just before it hits the wall, the ball is moving horizontally at an angle of 30° to the wall with speed u . The collision changes the ball's direction of movement through 45° . The diagram shows the situation viewed from above.



©2025 Exam Papers Practice. All rights reserved.

Model the ball as a particle.

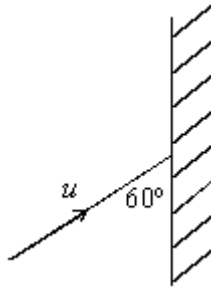
- (a) Show that the coefficient of restitution between the wall and the ball is approximately 0.464. (5)
- (b) Find the percentage of the kinetic energy which has been lost during the impact with the wall. (3)
- (c) Find the impulse, in terms of m and u , exerted on the ball by the wall. (3)
- (Total 11 marks)**

Q5.

A sphere of mass m , moving on a smooth horizontal surface, hits a smooth vertical wall. Just before it hits the wall, the sphere is moving at an angle of 60° to the wall with velocity

u .

The diagram shows the view from above.



The coefficient of restitution between the wall and the sphere is $\frac{3}{4}$.

- (a) Modelling the sphere as a particle, find the angle through which the direction of motion of the sphere is changed.
- (b) The impulse exerted by the wall on the sphere acts on the sphere for 0.05 seconds. Given that $m = 0.3$ kg, and $u = 5$ m s⁻¹ find the average impulsive force acting on the sphere.

(6)

(7)

(Total 13 marks)